

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Power Electronics	Course Code:	EE3032
	Program:	BS Electrical Engineering	Semester:	Spring 2025
	Assignment Submission Date:	12-Feb 2025	Total Marks:	50
	Section:	BEE-6A	Weight:	2
	Exam Type:	Assignment- 1	Page(s):	2
			CLO #	1

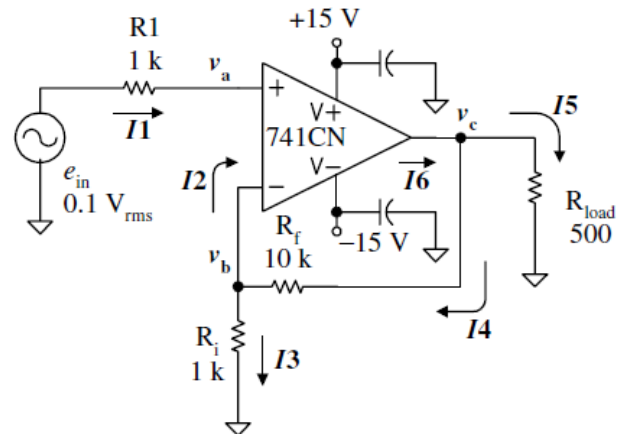
Student Name: _____ Roll No. _____ Section: _____

- Instructions:**
1. Use A4-sized sheets for this assignment.
 2. Avoid cutting and overwriting.
 3. Secure the pages with staple, clip, or binding. Do not submit loose pages.
 4. Assignment must be submitted on or before deadline during office hours only.
 5. Late submission will incur a penalty of 20% per day. After 2 days, assignment will no longer be accepted.

Question No. 1 (CLO No. 1)

Marks: 10

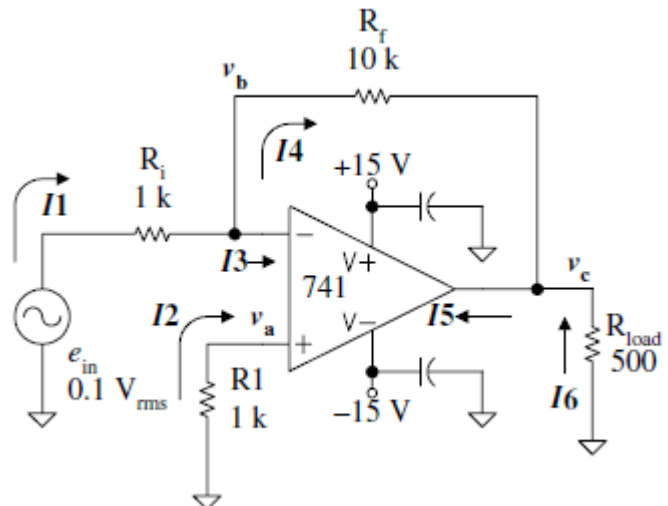
Analyze the op-amp circuit shown below and briefly explain its working. Also calculate all the indicated currents and voltages in the figure.



Question No. 2 (CLO No. 1)

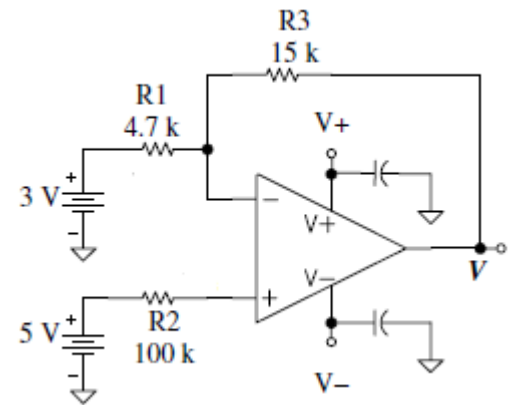
Marks: 10

Analyze the op-amp circuit shown below and briefly explain its working. Also calculate all the indicated currents and voltages in the figure.

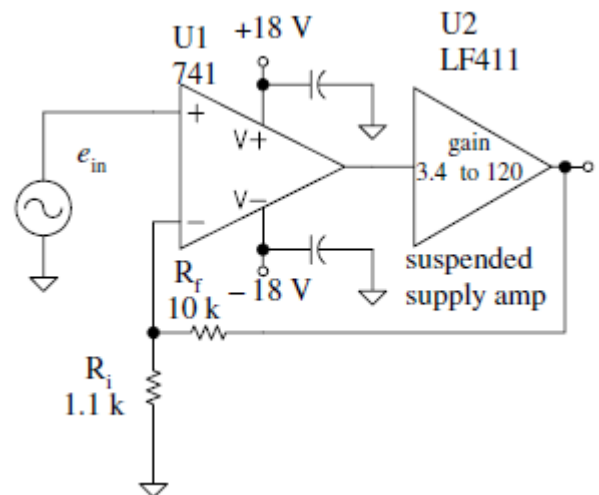


Question No. 3 (CLO No. 1)**Marks: 10**

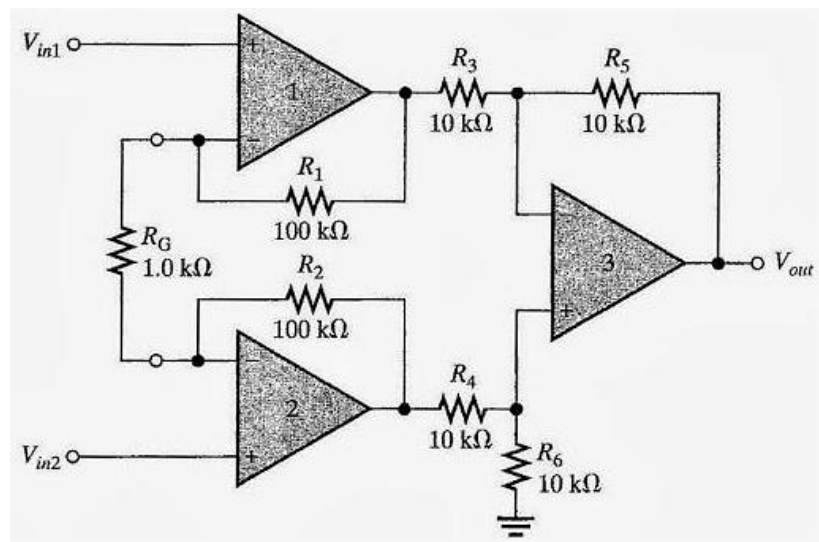
Analyze the op-amp application shown in the circuit below using Universal Analysis Technique.

**Question No. 4 (CLO No. 1)****Marks: 10**

Analyze the op-amp application shown in the circuit below using Universal Analysis Technique. Assume $e_{in} = 4.5$ V and a gain of 100 from the suspended supply amplifier.

**Question No. 5 (CLO No. 1)****Marks: 10**

Analyze the op-amp application shown in the circuit below and calculate the output voltage V_{out} . Consider $V_{in1} = 5$ mV and $V_{in2} = 10$ mV.



Power Electronics

Assignment #01

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22L-7638

BEE-6A

Q. No. 1

Given circuit is a non-inverting amplifier with gain

$$\Rightarrow G = 1 + \frac{R_f}{R_i}$$

$$\Rightarrow G = 1 + \frac{10k}{1k} = 11$$

$$\Rightarrow \frac{V_o}{V_i} = G = 11$$

$$V_o = 11(0.1)$$

$$\Rightarrow V_o = 1.1 \text{ Vrms}$$

$$\Rightarrow V_c = V_o = 1.1 \text{ Vrms}$$

$$V_+ \Rightarrow V_a = 0.1 \text{ Vrms}$$

$$V_- \Rightarrow V_b = V_a = 0.1 \text{ Vrms} \quad (\text{virtual short circuit})$$

$$\text{A } V = IR \Rightarrow I = \frac{V}{R}$$

$$\Rightarrow I_1 = \frac{V_a}{R_1}$$

$$\Rightarrow I_1 = \frac{0.1}{1k} = 0.1 \text{ mA} \quad \boxed{I_1 = 0.1 \text{ mA}}$$

⇒ Ideally, no current flows through $R_1 \Rightarrow I_1 = 0 \text{ A}$
As no current flows through op-amp,

$$\text{hence, } \Rightarrow \boxed{I_2 = 0 \text{ A}}$$

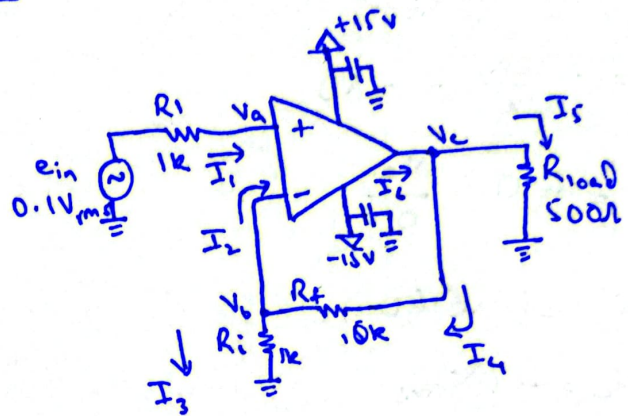
$$\Rightarrow I_3 = \frac{V_b}{R_i} = \frac{0.1}{1k} = 0.1 \text{ mA}$$

$$\boxed{I_3 = 0.1 \text{ mA}}$$

$$\Rightarrow V_c - V_b = I_4 R_f$$

$$\frac{1.1 - 0.1}{10k} = I_4$$

$$\Rightarrow \boxed{I_4 = 0.1 \text{ mA}}$$



$$\Rightarrow I_s = \frac{V_c}{R_l} = \frac{1.1}{500}$$

$$\boxed{I_s = 2.2 \text{ mA}}$$

$$\Rightarrow I_b = I_4 + I_s = 0.1 \text{ mA} + 2.2 \text{ mA}$$

$$\boxed{I_b = 2.3 \text{ mA}}$$

Q. No. 2

~~An inverting~~ inverting configuration
with gain

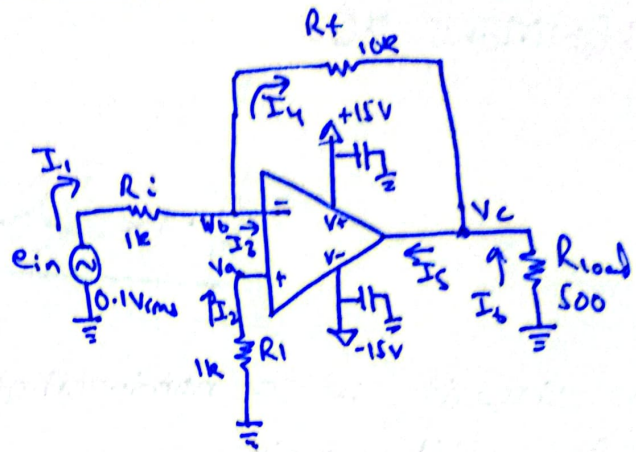
$$G = -\frac{R_f}{R_i}$$

$$\Rightarrow G = -\frac{10k}{1k} = -10$$

$$\Rightarrow \frac{V_o}{V_i} = -10 = \frac{V_c}{0.1}$$

$$\Rightarrow V_c = (-10)(0.1) =$$

$$V_c = -1 V_{rms}$$



A no current flows through op-amp (ideally),

$$\Rightarrow I_2 = 0$$

$$\Rightarrow I_7 = 0$$

$$\Rightarrow I_1 = \frac{0.1}{1k} = 0.1 \text{ mA}$$

$$I_1 = 0.1 \text{ mA}$$

$$\Rightarrow V_a = V_b = 0V \text{ (virtual short)}$$

$$\Rightarrow I_1 = I_4 = 0.1 \text{ mA}$$

$$\Rightarrow I_6 = \frac{V_c}{R_L} = \frac{-1}{500}$$

$$I_6 = -2 \text{ mA}$$

$$\Rightarrow \text{Ideally, } I_5 = 0A$$

$$\Rightarrow I_5 = I_4 + I_6$$

$$= 0.1 \text{ mA} + (-2 \text{ mA})$$

$$= -1.9 \text{ mA}$$

Q. 1.3

As no-current flows through non-inv terminal

$$\Rightarrow V_+ = 5V$$

$$\Rightarrow V_+ = V_- = 5V \quad (\text{Virtual short circuit})$$

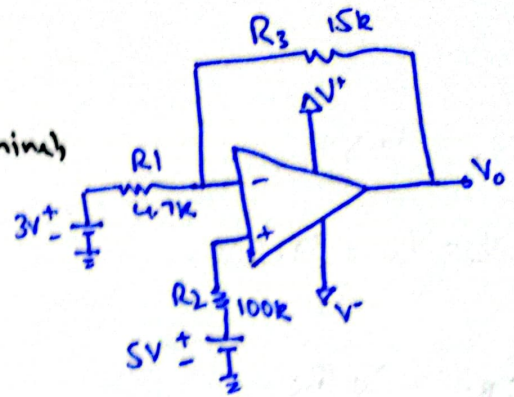
$$\Rightarrow I_{R1} = \frac{3-5}{4.7k} = -0.43mA$$

$$\Rightarrow I_{R2} = 0A$$

$$\Rightarrow I_{R3} = I_{R1} = 0.43mA$$

$$\begin{aligned} \Rightarrow V_{R3} &= I_{R3} R_3 \\ &= (0.43m)(15k) \\ &= 6.45V \end{aligned}$$

$$\begin{aligned} \Rightarrow V_o &= V_- + V_{R3} \\ &= 5 + 6.45 \\ &= 11.45V \end{aligned}$$



Q. No. 4

$$V_+ = 4.5V$$

$$\Rightarrow V_- = V_+ = 4.5V$$

$$\begin{aligned}\Rightarrow I_{R_i} &= V_- / R_i \\ &= (4.5) / (1.1k) \\ &= 4.09mA\end{aligned}$$

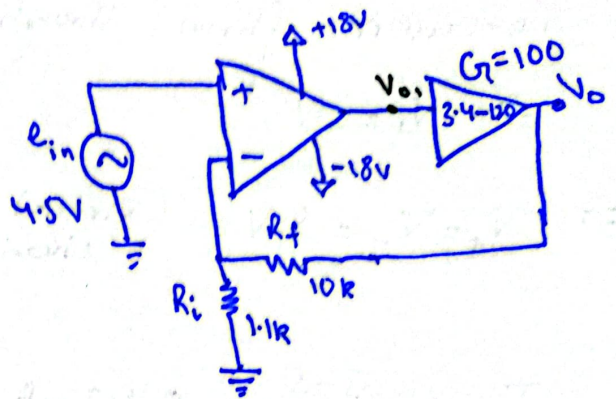
$$\Rightarrow I_{R_f} = I_{R_i} = 4.09mA$$

$$\begin{aligned}\Rightarrow V_{R_f} &= I_{R_f} R_f \\ &= (4.09mA)(10k) \\ &= 40.9V\end{aligned}$$

$$\begin{aligned}\Rightarrow V_{o_1} &= V_+ + V_{R_f} \\ &= 4.5 + 40.9 \\ &= 45.4V\end{aligned}$$

As supply is limited to $\pm 18V$,
hence, $V_{o_1} = 18V$

$$\begin{aligned}\Rightarrow V_o &= G(V_{o_1}) \\ &= (100)(18) \\ &= 1800V_o\end{aligned}$$



Q. 2.1 ~ 0.5

For an instrumentation amplifier,

$$\begin{aligned} V_o &= \left(\frac{R_5}{R_3} \right) \left(1 + \frac{2 R_1}{R_G} \right) (V_{in2} - V_{in1}) \\ &= \left(\frac{10K}{10K} \right) \left(1 + \frac{2(100K)}{1K} \right) (10m - 5m) \\ &= 1.005 V \end{aligned}$$

